

3 Measuring Spending Efficiency: Methodology and SACU Results

This section explains the stochastic-frontier methodology used in the Fiscal Monitor (IMF 2025f), reports the SACU-specific efficiency-gap estimates that enter the paper’s simulations, and discusses interpretation and caveats.

3.1 Methodology

The Fiscal Monitor methodology proceeds in three steps:

First, for each of four spending sectors—public investment, health, education, and research and development—a frontier is estimated via stochastic frontier analysis. The dependent variable is a sector-specific outcome measure (e.g., the infrastructure quality index for public investment, life expectancy at birth for health, learning-adjusted years of schooling for education). The explanatory variable is a five-year moving average of real sectoral spending per capita, controlled for country characteristics that affect outcomes independently of spending (population, income level, geography). The frontier represents the maximum outcome observed across countries with a given level of spending, net of factors outside the government’s short-run control.

Second, each country’s efficiency gap is measured as the distance from its own outcome to the frontier, expressed on a 0–1 scale in which 0 indicates full efficiency (the country is on the frontier) and 1 indicates full inefficiency. The five-year moving average smooths short-run noise and reflects the capital-stock-like nature of the spending–outcome relationship. For public investment, the Fiscal Monitor (IMF 2025g, Online Annex 1.2, footnote 9) notes that related studies such as Kapsoli et al. (2023) and Baum et al. (2020) typically use the public capital stock as the input measure, but reports that efficiency estimates do not change significantly when the capital stock is used in place of cumulative spending.

Third, the efficiency gaps feed directly into the DSGE model described in Section 5. An efficiency gap of 0.35 means that 35 percent of each unit spent on infrastructure “leaks”—through over-invoicing, poor project selection, weak implementation, or inadequate maintenance—and does not contribute to the productive public capital stock. The advantage of this formulation is that it separates the level of spending from the productivity of spending: a country with the same expenditure as its peers but a larger gap is wasting a higher share. Closing the gap raises productivity without requiring additional budget resources.

3.2 SACU Results

Efficiency gaps persist across sectors in SACU. Figure 3 reports the Fiscal Monitor efficiency gaps for the five SACU countries in three sectors—health, education, and infrastructure—at two points in time (2000 and 2023), alongside the unweighted averages for advanced economies, emerging markets, and sub-Saharan Africa. Three points stand out:

Infrastructure gaps have narrowed but remain wider than peers. The right panel of Figure 3 shows that all five SACU countries closed their infrastructure efficiency gaps substantially between 2000 and 2023, with Lesotho’s dramatic improvement from a gap close to 0.9 to about 0.4 being the most striking. Even so, the SACU average remains above the advanced-economy average and close to the emerging-market level. The progress reflects both rising spending and strengthened investment-management frameworks, but the remaining gap points to scope for further gains.

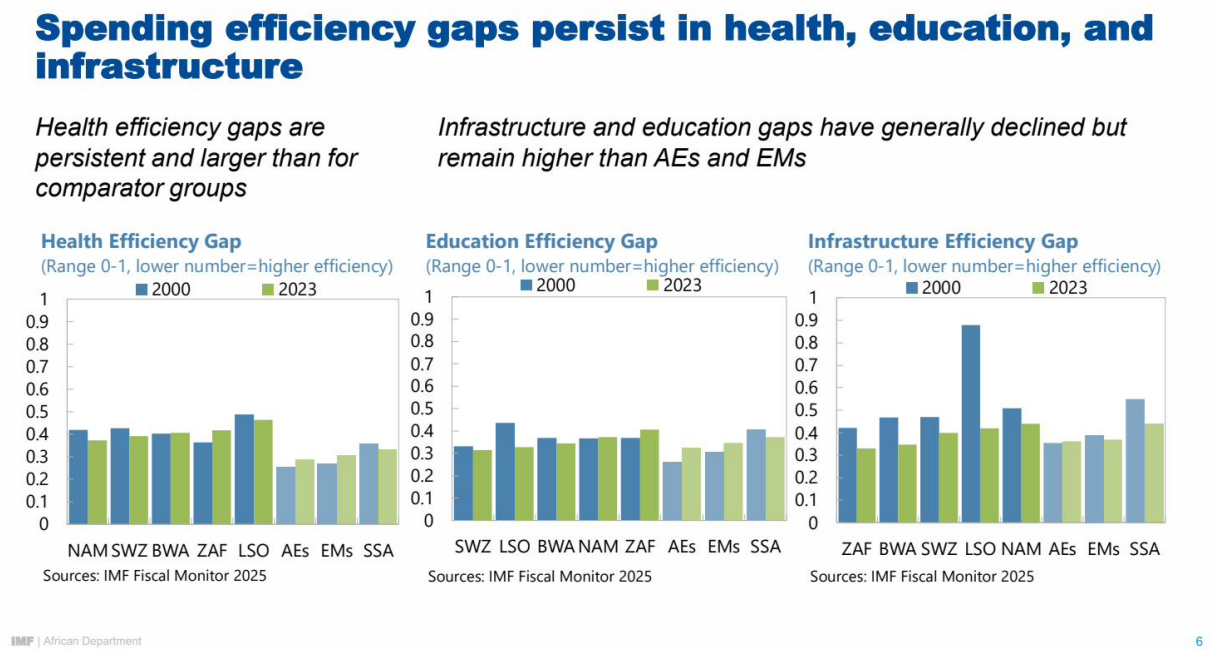
Education gaps have been broadly stable. The middle panel shows education efficiency gaps of 0.30 to 0.45 across SACU, with limited change between 2000 and 2023 and South Africa and Namibia modestly above their 2000 levels.

Health gaps are persistent and larger than peers. The left panel shows health efficiency gaps of 0.35 to 0.50 for SACU countries, little changed since 2000, and above the peer-group averages.

Health outcomes in SACU continue to reflect the lingering effects of the HIV/AIDS epidemic, but also structural features of the health systems—urban-rural coverage gaps, workforce pressures, and supply-chain weaknesses—that limit how efficiently spending translates into outcomes.

Figure 3: Efficiency Gaps in Health, Education, and Infrastructure Spending

Efficiency gaps, 0 to 1 scale. Higher values indicate larger gaps.



SOURCE: Reproduced from IMF (2026), based on IMF Fiscal Monitor (2025) efficiency-score database.

Infrastructure efficiency differences appear small but place SACU countries far apart in international comparison. Figure 4 sets the infrastructure gap in global context. It plots the 2023 infrastructure efficiency gap for every country in the Fiscal Monitor database, ordered from highest to lowest gap, with the five SACU countries highlighted. Namibia, Lesotho, and Eswatini sit close to the median of the global distribution; Botswana and South Africa are modestly below the median. None of the five SACU countries is among the best performers, confirming that the SACU distribution represents a set of economies with meaningful but not extreme scope for improvement.

Table 1 summarizes the country-specific efficiency gaps that feed the simulation exercise in Sections 5 and 6. The infrastructure gap is measured for the latest available year (2023). For human capital, we report both the education-specific gap (latest year 2024) and the health-specific gap (2023), as well as a blended gap defined as the simple average of the two. The blended gap is used in the model because the Fiscal Monitor DSGE framework treats education and health as a single public human-capital stock (schools and hospitals). The education-specific gap is reported separately because one of the reallocation variants targets education-sector investment only.

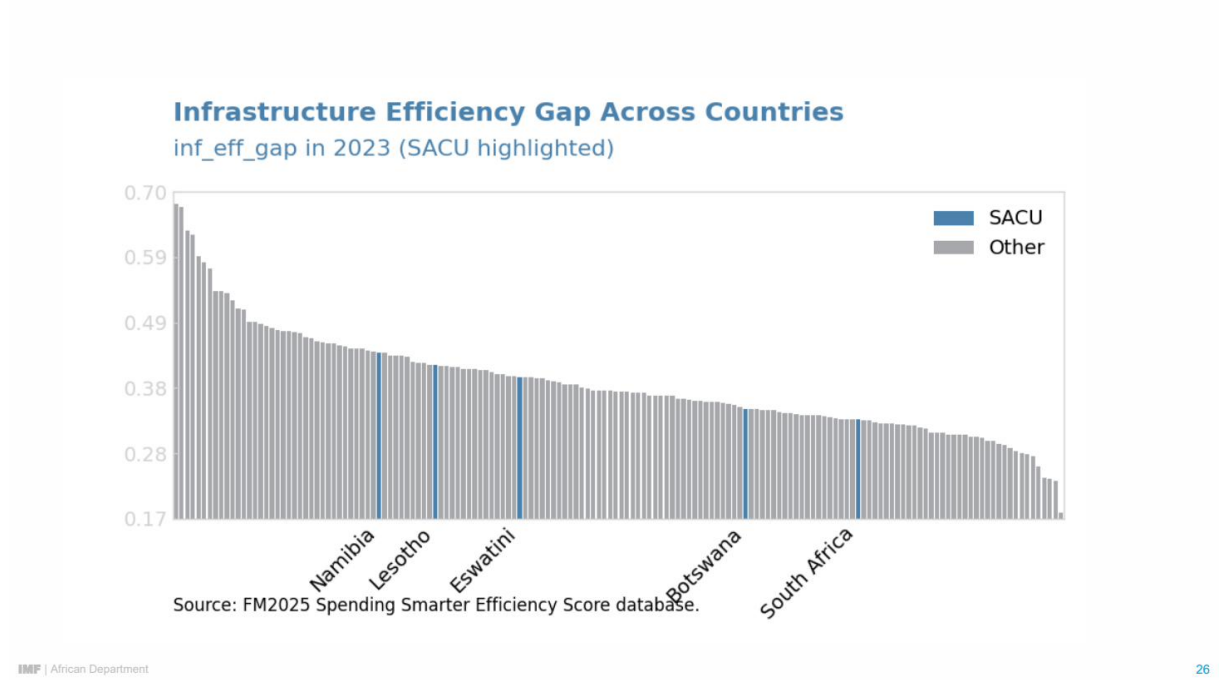
3.3 Interpretation and Caveats

Three interpretive caveats apply:

First, efficiency gaps conflate several underlying factors—institutional capacity, governance quality, the project mix, and country-specific shocks—and do not, on their own, identify the mechanism by which resources are lost. Section 4 links the measured gaps to institutional evidence from the Public Investment Management Assessments to address this.

Figure 4: Infrastructure Efficiency Gap Across Countries, 2023

Cross-country distribution of infrastructure efficiency gap, with SACU countries highlighted.



SOURCE: Reproduced from IMF (2026).

NOTE: Each bar is one country; SACU members shown in blue.

Table 1: SACU Efficiency Gaps, Latest Available Year

Efficiency gap, 0 to 1 scale; higher values indicate larger gaps. Used as inputs to the DSGE simulations.

Country	Infrastructure (e_{GI} , 2023)	Education (e_{GH}^{edu} , 2024)	Health (e_{GH}^{hlt} , 2023)	Blended HC (e_{GH}^{blend})
Botswana	0.34	0.38	0.40	0.39
Eswatini	0.39	0.35	0.41	0.38
Lesotho	0.40	0.42	0.47	0.45
Namibia	0.41	0.35	0.37	0.36
South Africa	0.33	0.37	0.37	0.37

SOURCE: IMF Fiscal Monitor (2025) efficiency-score database, extracted for the FM2025_SACU_DSGE model.

NOTE: Blended human-capital gap is the simple average of the education and health gaps.

Second, estimates are sensitive to the frontier specification. The Fiscal Monitor Online Annex (IMF 2025g) reports that alternative weightings of the underlying efficiency-score models can shift the level of estimated gaps by up to 26 percentage points in the extreme case. We use the baseline specification throughout, with sensitivity analysis discussed in Section 6.

Third, there are data-window limitations. The SFA uses a five-year moving average, which means the 2023 estimate reflects spending and outcomes between 2019 and 2023—a period disrupted by the COVID-19 shock, the subsequent SACU revenue boom, and commodity-price swings that affected Botswana in particular. Estimates should therefore be interpreted as reflecting the medium-run institutional capacity of each system rather than a precise snapshot of any single year.